

Lab_Day04-Securing Networks

Lab Objectives

By the end of this lab session, students will be able to:

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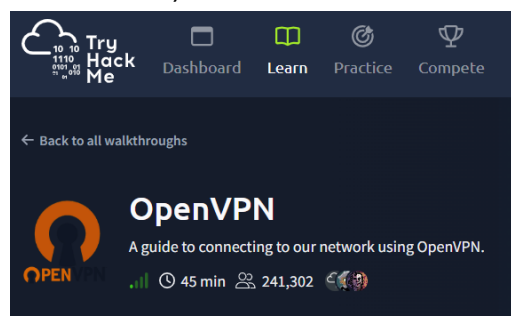
Lab Tasks

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Task 1: Using VPN Connection

Use [THM OpenVPN room](#) to establish VPN connection between your Kali Linux attack box and TryHackMe network to access target machines. This room will guide you about connection process. Use relevant connection guide to connect your machine to THM machines for future exercises.

From you Kali Linux, visit: <https://tryhackme.com/room/openvpn> and complete to room (CAPTURE YOUR FLAG).



Task 2: Investigati ng Network Traffic using Wireshark

Wireshark Tutorial: <https://www.youtube.com/watch?v=oSwipo6zSJ4>

1. Install Wireshark
2. Download Exercise_One.pcap file.

Exercise One

Open “Wireshark”, then use the “File” menu and the “Open” command to open the file “Exercise One.pcap”. You should see 26 packets listed.

This set of packets describes a ‘conversation’ between a user’s client and a central server. This entire conversation happens automatically, after a user types something and hits enter. Look at the packets to answer the following questions in relation to this conversation.

In answering the following questions, use brief descriptions. For example, “In frame X, the client requests a web page, and in frame Y, the server delivers the content of the page.”

Review: The TCP/IP network does not know how to route using names, such as www.yahoo.com. It only knows how to route using IP addresses. Therefore, a common name (CNAME), such as www.yahoo.com must be translated to an IP address, like 216.109.117.106 before your computer can request a web page.

Your computer uses a DNS request to lookup a CNAME and it gets back a set of IP addresses (a primary address and one or more backup addresses) that can be used to contact the server.

Note: Answers in white font colour.

a) What is the IP address of the client that initiates the conversation?





b) Use the first two packets to identify the server that is going to be contacted. List the common name, and three IP addresses that can be used for the server.

c) What is happening in frames 3, 4, and 5?

d) What is happening in frames 6 and 7?

e) Ignore frame eight. However, for your information, frame eight is used to manage flow control.

f) What is happening in frames nine and ten? How are these two frames related?

g) What happens in packet 11?

h) After the initial set of packets is received, the client sends out a new request in packet 12. This occurs automatically without any action by the user. Why does this occur?

i) What is occurring in packets 13 through 22?

j) Explain what happens in packets 23 through 26.

k) In one sentence describe what the user was doing (Reading email? Accessing a web page? FTP? Other?).

